

Bureaucratization as a Bumpy Journey: The Rise and Fall of the Imperial Chinese State

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Abstract

Historians have long documented the rise and fall of the Imperial Chinese state. This paper introduces bureaucratization as a potential mechanism underlying these processes. I model the principal-agent problem inherent in state administration and identify two groups of agents available to the ruler: elites and bureaucrats. Bureaucrats deliver higher returns to the ruler than elites because they can be monitored more effectively. Socioeconomic and technological developments allow the ruler to monitor an increasing share of administrative tasks and reassign them from elites to bureaucrats. Nonetheless, as bureaucrats gradually assume administrative duties, state capacity follows an N-shaped trajectory. It first rises and then declines because expanding opportunities for bureaucratic delegation eventually weaken the credibility of the ruler's relational contract with elites, inducing them to withdraw effort and reducing the ruler's aggregate payoff. Historically, this decline aligns with the major downturn experienced by the late imperial Chinese state. The model also predicts a resurgence of the modern Chinese state as bureaucratization reaches a more complete form. I further discuss the potential applicability of the model to the historical development of European states.

1. INTRODUCTION

Modern bureaucracy is commonly associated with the rise of eighteenth-century European fiscal-military states (Brewer, 1989; Storrs, ed, 2009). Interstate competition compelled rulers to develop bureaucratic administrations capable of extracting revenue and mobilizing mass armies for war (Tilly, 1990). The emerging bureaucracies are widely regarded as effective. Weber viewed the Prussian bureaucracy as an expression of Western rationalization and as technically superior to feudal rule (Weber, 1978).¹ Scholars also link growing state capacity in early modern Europe to bureaucratic expansion, emphasizing its role in imperial projection, social control, infrastructure provision, and economic coordination (MacDonagh, 1977).

However, bureaucracies long predated the modern state and were no panacea for political survival. Ancient civilizations with proto-bureaucracies, including Sumerian Mesopotamia, pharaonic Egypt, and Achaemenid Persia, ultimately collapsed (Scott, 2017; Tainter, 1988). Nor did bureaucratization prevent the disintegration of later empires such as the Byzantine Empire, the Ottoman Empire, and the Mughal Empire (Haldon, 1990; İnalçık, 1973; Richards, 1993). Imperial China provides perhaps the most prominent example. Although it developed a sophisticated bureaucracy and reached the height of its historical development at an early stage (Brandt et al., 1989; Elvin, 1973; Gernet, 1972), the late Imperial Ming and Qing states suffered from weak fiscal and military capacity (Huang, 1974; Rowe, 2009; Spence, 1990; Swope, 2014). Many scholars attribute this reversal to the late imperial bureaucracy, portraying it as rigid, corrupt, stagnant, and excessively oriented toward classical learning (Balazs, 1967; Elman, 2000; Huang, 1982; Kuhn, 1990; Lin, 2008; Ni and Van, 2006; Perkins, 1967; Zelin, 1984).

Does bureaucratization necessarily strengthen state capacity, and if so, why do its effects vary across time and place? Across Europe and Asia, political order was long organized through elite intermediaries who exercised considerable autonomy in local administration. Over time, many rulers sought to replace these intermediaries with bureaucracies (Eisenstadt, 1963; Fukuyama, 2011; Tilly, 1990; Weber, 1978). I define bureaucratization as the transfer of administrative tasks from elites to bureaucrats. Existing explanations focus on bureaucratic quality, most notably Weber’s distinction between rational and patrimonial bureaucracy. I

¹Weber (1978): “The decisive reason for the advance of bureaucratic organization has always been its purely technical superiority over any other form of organization. The fully developed bureaucratic apparatus compares with other organizations exactly as does the machine with the non-mechanical modes of production. Precision, speed, unambiguity, knowledge of files, continuity, discretion, unity, strict subordination, reduction of friction and of material and personal costs—these are raised to the optimum point in strictly bureaucratic administration. As compared with all collegiate, honorific, and avocational forms of administration, trained bureaucracy is superior on all the points.”

instead argue that bureaucratization itself generates a non-monotonic path of state capacity.

This paper develops a principal-agent model showing that even the transition to an ideal bureaucracy can generate an N-shaped trajectory of state capacity. In the model, the ruler delegates administrative tasks to two agents: elites and bureaucrats. Bureaucratic effort is perfectly observable, while elite effort is not. Initially, all tasks are delegated to elites. As society develops, bureaucrats become capable of performing a growing share of tasks, prompting the ruler to transfer tasks from elites to bureaucrats. This process generates an N-shaped trajectory of state capacity: rise, decline, and resurgence. State capacity initially rises because bureaucratic delegation reduces moral hazard. As more tasks become delegable to bureaucrats, however, the ruler becomes less dependent on elites and less willing to honor agreements with them. Anticipating this, elites reduce their effort. State capacity falls once declining elite effort outweighs the gains from bureaucratic expansion. State capacity recovers only when bureaucrats can perform a sufficiently broad range of tasks to offset the resulting loss of elite effort.

The mechanism operates through commitment problems in historical ruler-agent relations. In the absence of third-party enforcement, kings and emperors often reneged on agreements with their subordinates when expedient (Myerson, 2008, 2015; North and Weingast, 1989). Subordinates, in turn, could discipline rulers by withholding effort in future service. Cooperation was therefore sustained through self-enforcing relational contracts rather than formal enforcement. Rulers honored agreements not because they were externally constrained, but because doing so preserved the value of future cooperation. Accordingly, I model ruler-agent relations as repeated games under limited commitment. A central implication is that the expansion of bureaucratic institutions gradually alters the ruler's outside options vis-à-vis elites. When bureaucracy first emerges, the ruler's commitment to elites is largely unaffected. As bureaucracy develops, however, they become increasingly viable substitutes for elites. This raises the ruler's reservation payoff after a breakdown in cooperation with elites, thereby tightening the commitment constraint on elite effort. Only when bureaucrats can perform most administrative tasks does the ruler cease to depend on elite delegation, rendering the commitment problem vis-à-vis elites irrelevant.

The theory implies that bureaucratization need not increase state capacity monotonically. Instead, states may experience periods of strengthening, weakening, and recovery as bureaucratic institutions expand. Imperial China may have entered this process earlier than Europe because it bureaucratized sooner and more extensively. By the late imperial period, China may already have passed its initial peak and entered a period of relative decline, whereas many European states were only beginning their ascent.

This paper is situated within the formal literature on bureaucratic delegation. A common assumption in this literature is that principals possess authority over bureaucratic agents and a variety of institutional instruments for monitoring and incentivizing their behavior (Epstein and O'Halloran, 1994, 1999; Huber and Shipan, 2002; Lewis, 2008; McCubbins et al., 1987). A central concern is how principals select among alternative agents. One strand examines selection among bureaucrats with different hidden types, generating trade-offs such as ideological alignment versus information acquisition in democracies (Gailmard and Patty, 2007, 2012) and loyalty versus competence in autocracies (Egorov and Sonin, 2011; Zakharov, 2016). Related work examines delegation across different categories of agents, including politicians and bureaucrats (Alesina and Tabellini, 2007, 2008), politicians and judges (Maskin and Tirole, 2004), and agencies and courts (Stephenson, 2005). These agents differ along observable dimensions such as expertise, accountability, and ideological stability. Similarly, this paper examines delegation between elites and bureaucrats, who differ in their susceptibility to moral hazard and the range of tasks they can perform.

A key difference from existing models is that they take the commitment environment as given, whereas I show that the availability of alternative agents can itself reshape the ruler's commitment constraints. This difference stems from the move from static to repeated interactions. Existing models typically analyze agent selection in static games with perfect principal commitment. By contrast, the key trade-off here only arises in repeated interactions when commitment is self-enforcing. Because both delegation relationships coexist and generate value over time, a higher level of bureaucratic delegation simultaneously weakens the ruler's commitment to elites, leading to a decline in state capacity. This decline would not arise under a one-shot substitution.

The tension between alleviating agent moral hazard and sustaining ruler commitment is a novel feature of this paper. Prior work has examined the principal's trade-offs between alleviating moral hazard and other objectives, such as adverse selection and risk sharing (Grossman and Hart, 1992; Holmstrom, 1979; Holmstrom and Milgrom, 1991; Laffont and Tirole, 1993). In these models, principals optimally tolerate some degree of moral hazard because further reductions would come at the expense of other objectives. A similar logic emerges here. While bureaucratic delegation reduces agent moral hazard, it also weakens the ruler's commitment to elites. As a result, the ruler may be better off maintaining greater reliance on elites than would be optimal if moral hazard were the only concern. Under some parameter values, strategic interactions among rulers, elites, and bureaucrats push bureaucratization beyond the ruler's preferred level.

Limited principal commitment receives substantial attention in political science because political agreements often lack external enforcement. Gailmard (2009) discusses the inability

of the U.S. legislature to commit to state-contingent incentive schemes (“menu laws”) and its implications. The problem is even more acute in weak institutional environments such as authoritarian and historical settings (Levin, 2003). Myerson (2008, 2015) argue that rulers’ commitment problems in rewarding supporters are central to autocratic rule and that institutions such as courts can coordinate punishment and thereby enhance political survival. Following the seminal work of Baker et al. (1994), a growing literature shows how effective formal contracts can undermine relational agreements over the course of historical development. In financial history, relational borrowing is endogenously replaced by formal contracting as capital accumulation lowers interest rates and reduces the value of relational exchange (Ishiguro, 2016). In transitional economies, rulers’ relational commitments to the private sector can be undermined by a viable state sector (Aldashev et al., 2023). Building on these works, I show that the emergence of a new relationship can weaken the ruler’s commitment to an existing one and thereby reduce state capacity.

This paper also contributes to one of the most enduring debates in comparative history: the Great Divergence between China and Europe. The focus is on the state’s internal organization and its implications for state capacity, rather than on the wider determinants of economic and social development. In this respect, the paper speaks to a large literature that attributes the relative decline of Imperial China to its early political unification and centralization and the institutional consequences that followed. To begin, the large scale of the state makes it difficult to monitor agents effectively. Moreover, a strong state may lack commitment to its agents, as it can always expropriate their output at will. However, my model shows that these expropriatory incentives may not hold in a repeated setting, where the ruler has long-run incentives to sustain future cooperation with agents; instead, commitment problems may emerge elsewhere. Specifically, the negative effects are most severe when a new agent becomes viable but not yet sufficiently capable of fully replacing the old one, and less so when one agent clearly dominates the other. The ruler’s commitment is therefore endogenous and variable rather than fixed, causing state capacity to fluctuate over time.

2. HISTORICAL BACKGROUND

China bureaucratized earlier, faster, and more extensively than European states in history. Bureaucracy first appeared in China under the Zhou dynasty (ca. 1046–256 BCE), developed under the Qin and Han dynasties (221 BCE–220 CE), consolidated during the Sui and Tang dynasties (581–907 CE), and became the prevailing mode of administration from the Song dynasty (960–1279 CE) onward (Elman, 2000; Elvin, 1973; Hucker, 1975).

The Tang–Song transition is often understood as a shift from aristocratic-dominated to bureaucratic-dominated rule in Chinese historiography (Bol, 1994; Hartwell, 1982; Miyakawa, 1955). In Europe, comparable bureaucratic structures emerged only in the sixteenth century and flourished from the seventeenth century onward with the rise of fiscal-military states.

The earlier emergence of bureaucratic governance in China was rooted in socioeconomic conditions that differed substantially from those of Europe. The Chinese polity achieved early political unification and sustained central authority, which supported continuous administrative and technological development (Elvin, 1973; Hucker, 1975; Rosenthal and Wong, 2011). China attained an unusually high level of administrative maturity and technological sophistication early in its history, which likely facilitated the transition to bureaucratic rule. In contrast, the fall of the Roman Empire produced long-term political fragmentation in Western Europe (Hoffman, 2015; Scheidel, 2019). Roman administrative institutions and technologies were largely lost during the subsequent barbarian invasions and the feudalization of early medieval Europe (Epstein, 2009; Wickham, 2005). For centuries, European rulers relied primarily on feudal lords and other elite intermediaries for local governance, taxation, and military mobilization.

The shift from elite to bureaucratic governance reflected changes in the underlying technology of administration. Across early civilizations, rulers governed large territories primarily through elite intermediaries. This form of state organization was efficient when communication and transportation technologies were too primitive to support regular coordination between central and local authorities. Elite intermediaries were usually entrusted with most, if not all, local administrative functions, which depended largely on their local knowledge and discretionary judgment. Moreover, they often held hereditary status and shared power with rulers, allowing them to resist central oversight and extract informational rents. Precisely because these rents were valuable, elites had incentives to exert effort even without close monitoring. Unlike elites, bureaucrats derived authority from office rather than lineage. Their effectiveness depended largely on central monitoring: under perfect monitoring, they could be fully incentivized and would require no rents, whereas without monitoring, they might exert little effort. Advances in administrative technology, including paper, writing, postal systems, railways, telegraphs, and computers, as well as administrative institutions such as the Censorate, land registration, and royal courts, improved central-local coordination and expanded the range of activities subject to effective oversight. These developments enabled rulers to administer localities directly through bureaucrats.

The long-run consequences of bureaucratization for imperial state capacity are

ambiguous. Although China achieved remarkable state capacity in the early imperial period, expanding from a small polity in the Yellow River basin into one of the dominant powers on the Eurasian continent, many scholars argue that its state capacity declined relative to that of contemporary European powers during the late imperial era. By the late Qing, taxation remained low, the military was large but ineffective, and the state repeatedly failed to defend against foreign interventions and internal rebellions. The Qing state also fell short of the resources needed to foster industrialization comparable to that of contemporary Europe (Lovell, 2011; Platt, 2018; Zhang, 2023). Meanwhile, European fiscal-military states embarked on a sustained rise, increasing their fiscal capacity and projecting power on a global scale (Hoffman, 2015; Scheidel, 2019).

Despite extensive criticism in the literature, a closer examination of the late imperial Chinese bureaucracy suggests that its effectiveness varied systematically across tasks. It performed well in traditional tasks characteristic of an agrarian society, such as maintaining agricultural production, social order, and infrastructure including dams and canals (Brandt et al., 1989; Pomeranz, 2000; Rosenthal and Wong, 2011). Public welfare systems, such as the granary system, famine relief programs, and price stabilization mechanisms, functioned effectively for centuries (Li, 2007; Will, 1990; Will et al., 1991). The rules governing these systems were also meticulously maintained and periodically revised. However, it was less effective at addressing the novel challenges posed by the modern era, particularly industrial and military modernization. The Qing bureaucracy poorly managed the industrialization projects of the Self-Strengthening Movement and military conflicts with European powers. In many cases, the difficulty was not merely bureaucratic implementation but uncertainty about which policies were appropriate for an unfamiliar and rapidly changing environment. In general, bureaucracies tend to perform well on tasks governed by established procedures and effective oversight, but less well when confronting novel challenges for which administrative routines are lacking.

Yet another challenge for the late imperial Chinese state was the withdrawal of elites from meaningful political and military contributions. With the support of a strong bureaucracy capable of managing routine governance, late imperial rulers systematically curtailed elite power. These elites were often military aristocrats who had played central roles in founding their dynasties. Yet once new dynasties were consolidated, elites experienced peaceful displacement under the Song, violent purges under the Ming, and gradual marginalization under the Qing. Over time, scholar-officials came to dominate most civil administration and increasingly even military affairs. Elite effort and initiative consequently declined (Lovell, 2011; Platt, 2018; Swope, 2014). Historical accounts frequently portray late Qing elites, namely Manchu bannermen, as corrupt, ineffective, and

poorly motivated. They played a limited and often reactive role during the late Qing reforms (Lovell, 2011; Platt, 2018; Swope, 2014). In contrast, British elites were widely regarded by contemporaries and later commentators as competent, disciplined, and public-spirited (Beckett, 1986). They occupied leading positions in military command and used their political influence to advance the institutional and policy innovations underlying Britain’s fiscal, commercial, and industrial transformation (Holmes, 2008; Muir, 2013).

The late imperial Chinese state therefore faced great difficulties in modernizing its military and economy, with elites withdrawing effort and bureaucrats proving less effective at these tasks. However, the eventual transition to the modern era fundamentally altered this dynamic. The proliferation of new technologies has transformed many administrative tasks that once depended on discretionary, elite-based judgment into routinized and monitorable ones, enabling unprecedented levels of bureaucratic control. A prominent example is modern industrialization. Rulers now possess a wide informational toolkit, including GDP and employment statistics, tax records, satellite imagery, and social media traffic, readily available for monitoring and incentivizing bureaucratic effort (Li and Zhou, 2005; Maskin et al., 2000; Xu, 2011). Moreover, the CCP’s nomenklatura system has incorporated virtually the entire political system into the bureaucratic hierarchy. Aided by technologies such as computers, surveillance cameras, and AI tools, bureaucratic governance now extends deeply into economic and social life. Chinese state capacity appears to have rebounded (Ang, 2016; Brandt and Rawski, 2008; Naughton, 2007, 2018).

3. ADMINISTRATIVE ROUTINIZATION AND MODEL STRUCTURE

Before presenting the model, I introduce the concept of administrative routinization. The historical discussion indicates that certain socioeconomic preconditions must be met before rulers can shift administrative tasks from elites to bureaucrats. These include a sufficient degree of central authority and technological sophistication, conditions that often lie beyond the immediate control of rulers (Headrick, 1981; Porter, 1986, 1995; Standage, 1998). I summarize these preconditions in a single exogenous parameter, *administrative routinization*, drawing on Weber’s notion of the “impersonal and functional routinization of administration” (Weber, 1978). In the model, routinization denotes the share of administrative tasks that can be effectively delegated to bureaucrats. Higher levels of routinization imply that a broader range of administrative tasks can be standardized, supervised, and delegated to systematically selected, trained, and salaried bureaucrats.

A substantial literature has examined processes closely related to *routinization*. Scott (2020) argues that technological and social developments have made societies increasingly

“legible” to rulers by rendering land and populations more standardized, classifiable, and administratively visible. He cautions that greater legibility can facilitate high-modernist projects of social engineering and centralized control, many of which have ended disastrously. [Gailmard \(2023\)](#) shows that in the British American colonies, colonial autonomy was initially valuable for incentivizing colonists to discover profitable modes of production under uncertain resource endowments. As economic activities became more predictable, however, the English Crown increasingly pursued more direct forms of imperial regulation. More recently, a growing literature has examined how advances in information technology and artificial intelligence transform activities that once depended on tacit expertise into standardized procedures, allowing organizations to delegate them to less autonomous workers and to expand hierarchical systems of coordination and control ([Autor et al., 2003](#); [Brynjolfsson et al., 2025](#); [Garicano, 2000](#)).

History is thus conceptualized as a sequence of repeated games indexed by exogenously determined levels of routinization. In each game, the ruler, the elite, and the bureaucrat interact repeatedly over time, making decisions under the prevailing level of routinization. In each period, an administrative task arrives; the ruler allocates the task to either agent, the selected agent chooses effort, and the ruler determines transfers after observing output. Following [Levin \(2003\)](#), such repeated games admit stationary equilibria, allowing for tractable comparative statics across different levels of routinization. Importantly, the ruler retains ultimate authority over task allocation. The model therefore distinguishes between routinization and bureaucratization. Routinization is an exogenous parameter determined by broader socioeconomic development, whereas bureaucratization is the share of tasks allocated to bureaucrats in equilibrium.

Crucially, the level of routinization determines the relative value of the two delegation relationships and thus the ruler’s incentives to sustain cooperation with each agent. A low level of routinization implies that only a limited set of tasks can be delegated to bureaucrats in future periods, reducing the continuation value of bureaucratic delegation relative to that of elite delegation. As a result, the ruler has weak incentives to maintain cooperation with bureaucrats, because she is tempted to renege on them and rely instead on elite delegation. Anticipating this, bureaucrats exert little effort, further reducing the value of bureaucratic delegation. Low routinization therefore renders bureaucratic delegation unsustainable and delays bureaucratization. Historically, early bureaucratic reforms were often fragile and prone to reversal, a pattern frequently observed during the early stages of state formation ([Fukuyama, 2011](#); [Spruyt, 1994](#); [Tainter, 1988](#)).

At moderately high levels of routinization, the logic reverses. Routine tasks arrive frequently enough in future periods that bureaucratic delegation becomes an attractive

outside option, tempting the ruler to renege on elites. Anticipating this, elite effort declines, potentially to zero. Because bureaucrats are still unable to perform all tasks and fully replace elite contributions, the ruler’s payoff falls. This mechanism may help explain the waning fortunes of several heavily bureaucratized empires, including the Byzantine Empire, the Ottoman Empire, the Mughal Empire, and the Ming (1368–1644) and Qing (1644–1911) dynasties (Haldon, 1990; İnalçık, 1973; Richards, 1993; Rowe, 2009; Swope, 2014). In these cases, rulers may have weakened elite incentives before bureaucrats were capable of fully substituting for elite contributions at the prevailing technological level. The ruler’s payoff begins to recover only once elite effort falls to zero and routinization rises further, enabling bureaucrats to perform a sufficiently broad range of tasks to offset the loss of elite effort. This final stage may be reflected in the eventual revival of the modern Chinese bureaucratic state (Andreas, 2009; Lee, 1991; Li, 2001; Nathan, 2003).

The result is an N-shaped pattern of state capacity over levels of routinization, preceded by a plateau. The plateau reflects the initial delay in the onset of bureaucratization. Between the two extremes, however, state capacity increases with routinization, and the increase may even be linear over some regions of the parameter space. Within this range, both routine and non-routine tasks arrive frequently enough in future periods that both delegation relationships remain sufficiently valuable. The ruler can therefore credibly commit to both agents and induce efficient effort from each. This may help explain the historical rise of early imperial China and early modern European states, both of which experienced growing state capacity alongside the rapid expansion of bureaucratic administration (Brewer, 1989; Lewis, 2007, 2009; Storrs, ed, 2009).

Three modeling choices are worth noting. First, the baseline model assumes the transition into an ideal bureaucracy with no moral hazard. Specifically, bureaucratic effort is perfectly observable, whereas elite effort is not. This provides the most demanding test of the theory because it excludes bureaucratic inefficiency as a potential source of decline, making any decline harder to generate. The fact that a decline nevertheless emerges suggests that weakening state capacity is due not merely to the idiosyncratic failings of particular bureaucracies, but to a broader dilemma associated with bureaucratic governance itself. An extension confirms that introducing bureaucratic moral hazard leaves the main result unchanged and only amplifies the decline. Second, I use the ruler’s payoff as the measure of state capacity. This choice is consistent with historical settings in which the polity was often treated as the ruler’s personal patrimony. However, the combined payoffs of the ruler and agents exhibit the same N-shaped trajectory. Third, I assume that bureaucrats produce zero output when assigned non-routine tasks, reflecting their limited ability to perform them effectively. However, even if bureaucrats can perform non-routine

tasks, effective monitoring in such contexts is difficult. As a result, bureaucrats must be compensated with moral-hazard rents and effectively become elites.

The remainder of the paper is organized as follows. Section 4 presents the basic setup of the model. Section 5 analyzes a benchmark case with perfect ruler commitment. Section 6 develops the full model with self-enforcing relational commitment.

4. THE MODEL

In a historical regime, the players are a *ruler* ($\mathcal{R}$, she), an *elite* ($\mathcal{E}$, he), and a *bureaucrat* ($\mathcal{B}$, it), each modeled as a strategic unitary actor interacting in an infinitely repeated game at $t = 0, 1, \dots$. At each t , the ruler faces a task characterized by $\theta_t \in \{0, 1\}$, where $\theta = 1$ indicates that the task is *routine*. The ruler assigns the task to either the elite or the bureaucrat, $a_t \in \{\mathcal{E}, \mathcal{B}\}$, who becomes the carrier of the task at t . The carrier chooses a level of effort $e_t \in [0, 1]$ in trying to accomplish the task at the cost of $e_t^2/2$.² The carrier may succeed or fail in accomplishing the task, represented by $y_t \in \{0, 1\}$, where $y_t = 1$ indicates success. Accomplishment of the task generates a unit of an infinitely divisible output at the ruler's disposal. The ruler chooses a transfer $w_t \in [0, y_t]$ to compensate the carrier and pockets the rest $y_t - w_t$.

The bureaucratic agent differs from the elite agent in two respects. First, elites are more flexible than bureaucrats. Elites can perform both routine and non-routine tasks, whereas bureaucrats can perform only routine tasks.³ Formally, I assume that the probability that an agent successfully completes a task of type θ_t is given by

$$\Pr(y_t = 1 \mid \theta_t, a_t, e_t) = (\theta_t + (1 - \theta_t)\mathbb{I}(a_t = \mathcal{E})) e_t.$$

When the task is routine, that is, $\theta_t = 1$, both agents' efforts count toward output; when the task is non-routine, that is, $\theta_t = 0$, only the elite's effort counts toward output. The ruler is thus indifferent between delegating non-routine tasks to bureaucrats and leaving them idle, since bureaucrats exert zero effort on these tasks. Second, the ruler can better monitor the bureaucracy than the elite. Formally, I assume that the bureaucracy's effort is observable to the ruler, while the elite's effort is hidden, so that only elites face moral hazard. Note that both assumptions are simplified to keep the analysis tractable and to isolate the model's main trade-off; relaxing them does not affect the main results. For example, one could allow

²The functional form is chosen for analytical convenience, and the results are robust to alternative specifications with convex cost functions.

³An alternative interpretation is that even if bureaucrats are assigned to non-routine tasks, such as exploring new resources in early settlement, the ruler may be unable to codify procedures or monitor their actions effectively. In that case, bureaucrats would gain autonomy and functionally resemble elites.

bureaucrats to face moral hazard as well, as long as it is less severe than that of elites⁴, or allow bureaucrats to have a positive probability of producing good output on non-routine tasks, as long as they are less effective than elites. Taken together, the substantive difference is a matter of degree: elites are more flexible but less closely monitored, whereas bureaucrats are more specialized but more tightly monitored. This characterization is largely consistent with historical evidence.

At each t , a task arrives, and it is routine with probability $\Pr(\theta_t = 1) = \beta \in (0, 1)$. The parameter β indicates the share of routine tasks that can be handled by bureaucrats and measures the regime's level of *routinization*. The timing of the stage game is as follows:

1. A task $\theta_t \in \{0, 1\}$ is drawn and publicly observed.
2. The ruler publicly chooses a carrier of the task, $a_t \in \{\mathcal{B}, \mathcal{E}\}$.
3. The carrier a_t privately chooses a level of effort to accomplish the task, $e_t \geq 0$, and the ruler observes e_t if and only if $a_t = \mathcal{B}$.
4. Whether or not the task is accomplished $y_t \in \{0, 1\}$ is drawn and observed by the ruler and the carrier a_t .
5. The ruler transfers $w_t \in [0, y_t]$ to the carrier, which is observed by the ruler and the carrier.

Players care only about consuming outputs. At each t , the ruler receives the flow payoff of

$$y_t - w_t,$$

the elite receives

$$\mathbb{I}(a_t = \mathcal{E}) \left(w_t - \frac{e_t^2}{2} \right),$$

and the bureaucrat receives

$$\mathbb{I}(a_t = \mathcal{B}) \left(w_t - \frac{e_t^2}{2} \right).$$

All players are forward-looking and have a common discount factor $\delta \in (0, 1)$.

⁴This is shown in the extension, where a more realistic assumption is adopted, namely that the ruler incurs higher costs of monitoring elites than of monitoring bureaucrats. Accordingly, bureaucrats also face moral hazard, but to a lesser extent than elites. The main results continue to hold.

The solution concept is *subgame perfect Nash equilibrium* (SPNE). Players can condition their strategies on their observable histories of play. At t ,

$$\begin{aligned} h_t &= (\theta_\tau, a_\tau, (y_\tau, w_\tau))_{\tau=0}^{t-1} \in H \\ h_t^\mathcal{E} &= (\theta_\tau, a_\tau, (y_\tau, w_\tau)\mathbb{I}(a_\tau = \mathcal{E}))_{\tau=0}^{t-1} \in H^\mathcal{E} \\ h_t^\mathcal{B} &= (\theta_\tau, a_\tau, (y_\tau, w_\tau)\mathbb{I}(a_\tau = \mathcal{B}))_{\tau=0}^{t-1} \in H^\mathcal{B} \end{aligned}$$

represent the ruler's, the elite's, and the bureaucracy's observable histories, respectively. The ruler's strategy $\sigma^\mathcal{R}$ specifies the carrier of the task a_t given h_t and θ_t , and the transfer w_t given h_t , θ_t , a_t , y_t , and, if $a_t = \mathcal{B}$, also e_t . The elite's strategy $\sigma^\mathcal{E}$ specifies the level of effort e_t given $h_t^\mathcal{E}$, θ_t , and that $a_t = \mathcal{E}$. The bureaucracy's strategy $\sigma^\mathcal{B}$ specifies the level of effort e_t given $h_t^\mathcal{B}$, θ_t , and that $a_t = \mathcal{B}$. A strategy profile $\sigma = (\sigma^\mathcal{R}, \sigma^\mathcal{E}, \sigma^\mathcal{B})$ is an equilibrium if each player's strategy is a perfect best response against the other players' strategies. I focus on characterizing the ruler-optimal equilibrium σ^* that maximizes the ruler's payoff at $t = 0$.

In what follows, the time subscript “ t ” is omitted whenever it causes no confusion, and I maintain the parametric assumption that $\delta > 3/4$.

Two modeling choices are worth noting. First, some actions are not publicly observable, so players condition their strategies only on their observable histories of play. Specifically, I assume that agents do not observe one another's interactions with the ruler—namely, their effort choices or the transfers they receive—and therefore cannot condition their strategies on such information. Instead, they can condition on actions that are either publicly observable or privately observed by themselves. Public observable actions include the allocation of tasks: if an agent is assigned a task that, under the equilibrium agreement, should be assigned to the other agent, this deviation is publicly observable and can trigger punishment. This assumption is made primarily for analytical tractability. Allowing agents to observe each other's interactions with the ruler and to coordinate punishments accordingly would substantially complicate the analysis without affecting the central intuition. The assumption is also historically plausible, as elites and bureaucrats often constituted distinct social groups embedded in separate institutional structures, making full informational transparency and coordination between the two groups unlikely. Second, I focus on the ruler-optimal equilibrium among all feasible equilibria. The technical reason is that the ruler moves first in the stage game, so she can select her payoff-maximizing equilibrium. Substantively, the central question is whether even this maximum payoff declines.

5. BENCHMARK: PERFECT COMMITMENT OF THE RULER

To build intuition, I begin by analyzing a benchmark in which the ruler can commit perfectly to transfers. In this setting, dynamic incentives are absent, and the problem reduces to a one-shot delegation decision.

First, consider the case in which the ruler assigns a routine or non-routine task to the elite, $a = \mathcal{E}$. Because the elite's effort is unobservable to the ruler, the ruler can condition the transfer only on whether the task is successfully accomplished. Formally, the ruler can commit to a transfer schedule $W(y)$ that specifies a transfer $W(1)$ when the elite succeeds, $y = 1$, and $W(0)$ when the elite fails, $y = 0$. Because $W(y) \in [0, y]$ must hold, it follows that $W(0) = 0$, so that the ruler's transfer schedule is fully characterized by $W(1) \in [0, 1]$. Given $W(1)$, the elite chooses a level of effort e that maximizes

$$eW(1) - \frac{e^2}{2},$$

which yields $\hat{e} = W(1)$. In turn, the ruler can induce any effort level $\hat{e}$ by setting $W(1) = \hat{e}$. Doing so, the ruler expects to get

$$\hat{e}(1 - \hat{e}),$$

which is maximized by setting $\hat{e} = 1/2$. Therefore, if the ruler assigns a routine or non-routine task to the elite, she commits to transfer $1/2$ upon success, and the elite exerts effort $1/2$, so that the task is successfully completed with probability $1/2$. The maximal payoff the ruler obtains from assigning any task to the elite is thus $1/4$.

Second, consider the case in which the ruler assigns a routine task to the bureaucrat, $a = \mathcal{B}$. In contrast to the case with the elite, the bureaucrat's effort is observable to the ruler. As a result, the ruler can condition the transfer on both whether the task is successfully accomplished, y , and on the bureaucrat's effort, e . Formally, the ruler can commit to a transfer schedule $W(y, e)$. Because $W(y, e) \in [0, y]$ must hold, $W(0, e) = 0$ for all $e \in [0, 1]$. By choosing $e = 0$, the bureaucrat secures a payoff of 0. Hence, for any given $\tilde{e} \in (0, 1]$, the ruler can induce $e = \tilde{e}$ by setting $W(1, e) = 0$ for all $e \neq \tilde{e}$ and $W(1, \tilde{e}) = \tilde{e}/2$, which ensures

$$\tilde{e}W(1, \tilde{e}) - \frac{\tilde{e}^2}{2} \geq 0.$$

Given that the ruler wants to induce $\tilde{e}$, it is optimal for the ruler to set $W(1, \tilde{e}) = \tilde{e}/2$. In

turn, the ruler chooses $\tilde{e} \in [0, 1]$ to maximize

$$\tilde{e} \left(1 - \frac{\tilde{e}}{2}\right),$$

which yields $\tilde{e} = 1$. Therefore, if the ruler assigns a routine task to the bureaucrat, she commits to transfer $1/2$ upon successful completion and verification of effort $e = 1$. The bureaucrat then exerts effort equal to 1, ensuring that the task is successfully accomplished. The maximal payoff the ruler obtains from assigning a routine task to the bureaucrat is thus $1/2$.

Comparing the two cases, the ruler is strictly better off assigning routine tasks to the bureaucrat than to the elite whenever feasible. Given that $\theta = 1$, choosing $a = \mathcal{B}$ yields a payoff of $1/2$ to the ruler, while choosing $a = \mathcal{E}$ yields $1/4$. Note that, regardless of the assignment, the carrier receives a transfer of $1/2$ upon accomplishing the task. Such a transfer induces only effort $\hat{e} = 1/2$ from the elite but $\tilde{e} = 1$ from the bureaucrat. This difference arises from the ruler's ability to monitor the bureaucrat's effort and her inability to observe that of the elite. Because the ruler can monitor the bureaucrat's effort, she can condition transfers on the required effort and punish deviations. By contrast, because the ruler cannot monitor the elite's effort, she can condition transfers only on whether the task is accomplished. This makes the elite's effort more expensive than that of the bureaucrat. As shown above, it costs the ruler $e/2$ to induce effort e from the bureaucrat, whereas inducing the same level of effort from the elite costs e .

Proposition 1 (Benchmark). *Assume the ruler has perfect commitment. In the ruler-optimal equilibrium,*

- *if the task is non-routine, $\theta = 0$, the ruler assigns the task to the elite, the elite exerts the effort of $1/2$, and the ruler makes the transfer of*

$$W^{\mathcal{E}}(y) = \begin{cases} \frac{1}{2}, & y = 1 \\ 0, & \text{otherwise} \end{cases};$$

- *if the task is routine, $\theta = 1$, the ruler assigns the task to the bureaucrat, the bureaucrat exerts the effort of 1, and the ruler makes the transfer of*

$$W^{\mathcal{B}}(y, e) = \begin{cases} \frac{1}{2}, & y = 1, e = 1 \\ 0, & \text{otherwise} \end{cases}.$$

In this ruler-optimal equilibrium, the bureaucrat is fully extracted. It receives a payoff of 0, and the ruler captures the entire value generated by bureaucratic effort. In contrast,

the elite retains a payoff of

$$\frac{1}{2} \frac{1}{2} - \frac{\left(\frac{1}{2}\right)^2}{2} = \frac{1}{8} > 0.$$

This payoff is the elite's *moral hazard rent* that the ruler must forgo due to her inability to monitor elite effort. The incentive schemes for elites and bureaucrats therefore differ. Elites receive moral hazard rents, which serve as part of the incentive mechanism sustaining their effort on routine and non-routine tasks. By contrast, bureaucrats receive no rents on routine tasks and are primarily incentivized by the ruler's monitoring. In this sense, even if bureaucrats were assigned non-routine tasks, they would earn rents and effectively become elites whenever monitoring or discipline is not feasible.

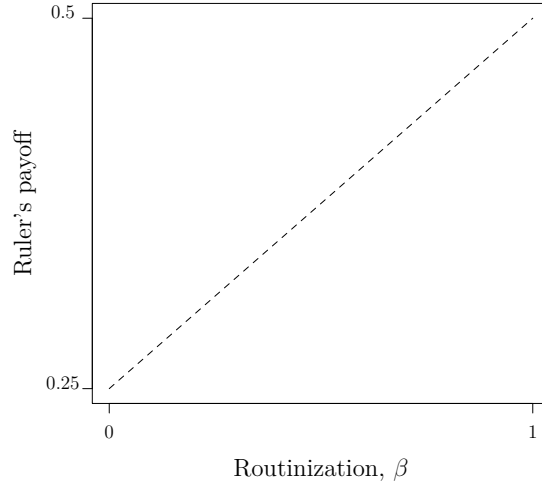
In summary, under perfect commitment, the ruler's maximal payoff is

$$(1 - \beta) \frac{1}{4} + \beta \frac{1}{2} = \frac{1 + \beta}{4},$$

which is increasing linearly in β . As routinization rises, the ruler gradually reallocates routinized tasks to the bureaucrat, while non-routine tasks remain with the elite, generating a smooth transition toward bureaucratization. This shift reduces the moral hazard rents paid to the elite on routine tasks and thereby increases the ruler's payoff in proportion to the degree of routinization. More generally, bureaucratization benefits the ruler by mitigating moral hazard among her agents. Figure 1 illustrates this benchmark, shown by the dashed line. Although the figure suggests an optimistic view of bureaucratization, historical evidence points to a more complex and contingent relationship. This contrast motivates relaxing the assumption of perfect ruler commitment, which rarely holds in historical settings.

6. THE RULER'S COMMITMENT PROBLEM

I now investigate the case in which the ruler cannot credibly commit to transfers to agents. Commitment must therefore be self-enforcing, supported by the threat that agents will terminate cooperation whenever the ruler fails to honor promised transfers. Following [Abreu \(1988\)](#) and [Levin \(2003\)](#), it is without loss of generality to focus on equilibria that are stationary on the equilibrium path and enforced by the most severe punishment off the equilibrium path. The most severe punishment maximizes the cost of deviation and supports the greatest level of commitment that the ruler can sustain. In this model, such punishment entails either one or both agents exerting zero effort in every period in which they are assigned a task. In this section, I first characterize the set of equilibria that can be



Dashed: perfect commitment; Solid: imperfect commitment.

Figure 1: Ruler's payoff as a function of routinization, β .

sustained by this punishment in bilateral relationships between the ruler and each agent. In the next section, I select the ruler-optimal equilibrium by comparing these sustainable equilibria.

6.1. Commitment to the bureaucrat

Suppose there exists an equilibrium in which the ruler assigns routine tasks to the bureaucrat and the bureaucrat exerts positive effort.⁵ As shown in the benchmark, this requires the ruler to sustain an equilibrium transfer scheme,

$$W(y, e) = \begin{cases} \tilde{e}, & y = 1, e = \tilde{e} \\ 0, & \text{otherwise} \end{cases},$$

where $\tilde{e} \in [0, 1]$ is the level of effort the ruler wants to induce from the bureaucrat.

If the ruler fails to make the transfer $W(1, \tilde{e})$ after the bureaucrat has exerted effort $\tilde{e}$ and successfully accomplished the routine task, $y = 1$, the most severe punishment that can be inflicted on the ruler has two components. First, in all future periods, whenever the ruler assigns a routine task to the bureaucrat, the bureaucrat exerts zero effort. Second, whenever the ruler assigns a routine task to the elite, the elite likewise exerts zero effort. Although the elite cannot directly observe the ruler's defection against the bureaucrat, he

⁵Non-routine tasks are irrelevant for the discussion here, since bureaucrats cannot perform non-routine tasks.

can infer this defection when the ruler assigns a routine task to him. The dual punishments eliminate all future cooperation on routine tasks and constitute the most severe sanctions available.⁶ It follows that the effectiveness of the punishments depends crucially on β , which determines the frequency of routine cooperation and the continuation value at stake under the punishments. Accordingly, I characterize the equilibria that can be sustained by the most severe punishments over different ranges of β .

To begin, given $\tilde{e}$, the ruler's per-period payoff from honoring the promised transfer $W(y, e)$ to the bureaucrat and thus retaining the option to assign routine tasks to the bureaucrat is

$$(1) \quad \beta \tilde{e} \left(1 - \frac{\tilde{e}}{2}\right).$$

Note that this payoff is strictly increasing in routinization, β . With a larger β , the ruler can assign routine tasks to the bureaucrat more frequently and thus gains more from sustaining cooperation with the bureaucrat in future periods.

Next, once the bureaucrat has exerted the required effort $\tilde{e}$ and accomplished the task, $y = 1$, the ruler decides whether to defect. By making the promised transfer $W(1, \tilde{e}) = \frac{\tilde{e}}{2}$, the ruler obtains from the bureaucrat a total discounted payoff from the current period onward of

$$(1 - \delta) \left(1 - \frac{\tilde{e}}{2}\right) + \delta \beta \tilde{e} \left(1 - \frac{\tilde{e}}{2}\right).$$

She receives 1 net of the transfer to the bureaucrat in the current period and, by keeping her promise, expects to receive the payoff given by (1) from future cooperation with the bureaucrat on routine tasks. Alternatively, by reneging on her promised transfer, the ruler obtains

$$1 - \delta$$

⁶The mechanism of dual punishment allows the elite or the bureaucrat to punish the ruler following her defection against the other agent. This mechanism arises directly from the construction of the most severe punishment under the model's information structure and is necessary to sustain the maximal level of commitment by the ruler. Formally, the contract among the three parties specifies task-allocation rules and also prescribes punishment by the non-assigned agent in the event of a deviation. This mechanism may be somewhat stylized, as it requires agents to be sufficiently forward-looking to discipline the ruler following her defection against the other agent in order to sustain maximal commitment. However, removing this mechanism, or restricting agents from punishing the ruler when she defects against the other agent, only makes the ruler's commitment to both agents more difficult to sustain overall. It would not alter the N-shaped pattern of the ruler's payoff but may affect the thresholds at which the kinks arise and shift the payoff function vertically.

as the total discounted payoff from the current period onward when the task is routine. Recall that once the ruler reneges on her promised transfer to the bureaucrat, both the bureaucrat and the elite exert zero effort on any assigned routine task forever thereafter. Therefore, the ruler is willing to keep her promise to make the transfer of $W(1, \tilde{e}) = \frac{\tilde{e}}{2}$ if and only if

$$(1 - \delta) \left(1 - \frac{\tilde{e}}{2}\right) + \delta\beta\tilde{e} \left(1 - \frac{\tilde{e}}{2}\right) \geq 1 - \delta,$$

which yields

$$\tilde{e} \leq 2 - \frac{1 - \delta}{\delta\beta}.$$

It follows that the ruler can induce the ruler-optimal level of bureaucratic effort, $\tilde{e}(\beta) = 1$, only if $2 - \frac{1 - \delta}{\delta\beta} \geq 1$, that is, if $\beta \geq \frac{1 - \delta}{\delta}$. When $2 - \frac{1 - \delta}{\delta\beta} \leq 1$, i.e., when $\beta \leq \frac{1 - \delta}{\delta}$, the maximal inducible bureaucratic effort is bounded above by $2 - \frac{1 - \delta}{\delta\beta}$. When $2 - \frac{1 - \delta}{\delta\beta} \leq 0$, i.e., when $\beta \leq \frac{1 - \delta}{2\delta}$, the ruler can induce only zero bureaucratic effort. Thus, when β is low, equilibrium bureaucratic effort is bounded above by the ruler's limited commitment. For low values of β , routine tasks occur infrequently, implying a low continuation value of cooperation with the bureaucrat on routine tasks and a weaker punishment following reneging. As a result, the ruler has a stronger incentive to renege on promised transfers after observing success ($y = 1$), which limits the transfers she can credibly promise. Anticipating this, the bureaucrat supplies a bounded level of effort, as higher effort would require transfers that are not credibly enforceable.

Lemma 1. *There exists an equilibrium in which the ruler assigns routine tasks to the bureaucrat and the bureaucrat exerts positive effort if and only if*

$$(2) \quad \beta > \frac{1 - \delta}{2\delta}.$$

Given that $\beta > \frac{1 - \delta}{2\delta}$, the maximal effort the ruler can induce from the bureaucrat in equilibrium is

$$(3) \quad \tilde{e}^*(\beta) := \min \left\{ 2 - \frac{1 - \delta}{\delta\beta}, 1 \right\}$$

and $\tilde{e}^(\beta) = 1$ if and only if*

$$(4) \quad \beta \geq \frac{1 - \delta}{\delta} > \frac{1 - \delta}{2\delta}.$$

Figure 2 provides a visual illustration for Lemma 1. As discussed above, in an equilibrium where the ruler delegates routine tasks to the bureaucrat, a higher β allows the ruler to commit to larger credible transfers and induce higher bureaucratic effort. Therefore, when $\beta \geq \frac{1-\delta}{\delta}$, the optimal effort level $\tilde{e}(\beta) = 1$ can be induced in equilibrium; when $\beta \in (\frac{1-\delta}{2\delta}, \frac{1-\delta}{\delta})$, a positive but inferior effort level $\tilde{e}(\beta) \in (0, 1)$ can be induced; and when $\beta \leq \frac{1-\delta}{2\delta}$, no positive effort can be induced.

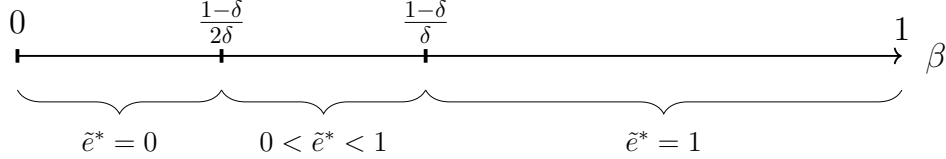


Figure 2: Bureaucratic effort as a function of routinization, β .

6.2. Commitment to the elite

In general, there are two possible kinds of equilibria in which the ruler may assign any task to the elite: one in which the ruler assigns all tasks to the elite, and the other in which the ruler assigns routine tasks to the bureaucrat and non-routine tasks to the elite. As shown in the benchmark, in either case, the ruler is expected to sustain an equilibrium transfer scheme to the elite,

$$W(y) = \begin{cases} \hat{e}, & y = 1 \\ 0, & \text{otherwise} \end{cases},$$

where $\hat{e} \in [0, 1]$ is the level of effort the ruler wants to induce from the elite.

Delegate All to the elite. Suppose there exists an equilibrium in which the ruler assigns all tasks to the elite and the elite exerts positive effort. The ruler's per-period payoff from honoring the promised transfer $W(y)$ to the elite and thus retaining the option to assign all tasks to the elite is

$$(5) \quad \hat{e}(1 - \hat{e}).$$

Once the elite has successfully completed the task, $y = 1$, the ruler decides whether to defect. By making the promised transfer $W(1) = \hat{e}$, the ruler obtains from the elite a total

discounted payoff from the current period onward of

$$(1 - \delta)(1 - \hat{e}) + \delta\hat{e}(1 - \hat{e}).$$

She receives 1 net of the transfer to the elite in the current period and, by keeping her promise, expects to receive the payoff given by (5) from future cooperation with the elite on all tasks. Alternatively, by reneging on her promised transfer, the ruler obtains

$$1 - \delta$$

as the total discounted payoff from the current period onward. Recall that if the ruler reneges on her promised transfer to the elite, she loses the continuation value of future cooperation with both agents on routine and non-routine tasks thereafter. Not only does the elite exert zero effort on any assigned task, but the bureaucrat also exerts zero effort on any assigned routine task. This is because, although the bureaucrat cannot directly observe the ruler's defection against the elite, the bureaucrat can infer this deviation when the ruler assigns a routine task to the bureaucrat. Therefore, the ruler is willing to keep her promise to make the transfer of $W(1) = \hat{e}$ if and only if

$$(1 - \delta)(1 - \hat{e}) + \delta\hat{e}(1 - \hat{e}) \geq 1 - \delta,$$

which yields

$$\hat{e} \leq \frac{2\delta - 1}{\delta}.$$

Note that the effort level that the ruler seeks to induce from the elite is $\hat{e}(\beta) = \frac{1}{2}$, which can be supported for all values of β , since $\frac{1}{2} < \frac{2}{3} \leq \frac{2\delta-1}{\delta}$. This is because delegating all tasks to the elite generates a sufficiently large continuation value of cooperation. The threat of losing this value disciplines the ruler, enabling her to commit to the transfer required to induce the ruler-optimal level of elite effort.

Lemma 2. *There always exists an equilibrium in which the ruler assigns all tasks to the elite and the elite exerts positive effort. Among these equilibria, there exists a unique one that maximizes the ruler payoff and in this equilibrium, the ruler induces the effort of 1/2 from the elite.*

Figure 3 provides a visual illustration for Lemma 2. As discussed above, in an equilibrium where the ruler delegates all tasks to the elite, the ruler can induce the ruler-optimal level of elite effort $\hat{e}(\beta) = \frac{1}{2}$ for all values of β .

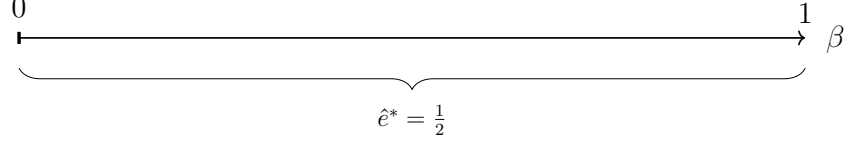


Figure 3: Elite effort as a function of routinization, β .

Delegate *Non-routine* to the elite. Suppose there exists an equilibrium in which the ruler assigns only non-routine tasks to the elite and the elite exerts positive effort. The ruler's per-period payoff from honoring the promised transfer $W(y)$ to the elite and thus retaining the option to assign non-routine tasks to the elite is

$$(6) \quad (1 - \beta)\hat{e}(1 - \hat{e}).$$

where $\hat{e} \in [0, 1]$ is the level of effort the ruler wants to induce from the elite.

Once the elite has successfully completed the task, $y = 1$, the ruler decides whether to defect. By making the promised transfer $W(1) = \hat{e}$, the ruler obtains from the elite a total discounted payoff from the current period onward of

$$(1 - \delta)(1 - \hat{e}) + \delta(1 - \beta)\hat{e}(1 - \hat{e}).$$

She receives 1 net of the transfer to the elite in the current period and, by keeping her promise, expects to receive the payoff given by (6) from future cooperation with the elite on non-routine tasks. Alternatively, by reneging on her promised transfer, the ruler obtains

$$1 - \delta$$

as the total discounted payoff from the current period onward when the task is non-routine. Recall that if the ruler reneges on her promised transfer to the elite, she loses the continuation value of future cooperation with the elite on non-routine tasks, as the elite exerts zero effort on any assigned non-routine task thereafter. Note that the bureaucrat always exerts zero effort on non-routine tasks by assumption, so the ruler never delegates such tasks to it. Therefore, the ruler is willing to keep her promise to make the transfer $W(1) = \hat{e}$ if and only if

$$(1 - \delta)(1 - \hat{e}) + \delta(1 - \beta)\hat{e}(1 - \hat{e}) \geq 1 - \delta,$$

which yields

$$\hat{e} \leq 1 - \frac{1 - \delta}{\delta(1 - \beta)}.$$

It follows that the ruler can induce the ruler-optimal level of elite effort, $\hat{e} = \frac{1}{2}$, only if $\frac{1}{2} \leq 1 - \frac{1 - \delta}{\delta(1 - \beta)}$, that is, if $\beta \leq \frac{3\delta - 2}{\delta}$. When $\frac{3\delta - 2}{\delta} < \beta \leq 1$, the maximal inducible elite effort is bounded above by $1 - \frac{1 - \delta}{\delta(1 - \beta)}$. When $1 - \frac{1 - \delta}{\delta(1 - \beta)} \leq 0$, that is, when $\beta \geq \frac{2\delta - 1}{\delta}$, the ruler can induce only zero elite effort. Thus, when β is high, equilibrium elite effort is bounded above by the ruler's limited commitment. For high values of β , non-routine tasks occur infrequently, implying a low continuation value of cooperation with the elite on non-routine tasks and a weaker punishment following renegeing. As a result, the ruler has a stronger incentive to renege on promised transfers after observing success ($y = 1$), which limits the transfers she can credibly promise. Anticipating this, the elite supplies a bounded level of effort, as higher effort would require transfers that are not credibly enforceable.

Lemma 3. *There exists an equilibrium in which the ruler assigns non-routine tasks to the elite, and the elite exerts positive effort if and only if*

$$(7) \quad \beta < \frac{2\delta - 1}{\delta}.$$

Given that $\beta < \frac{2\delta - 1}{\delta}$, the optimal effort the ruler can induce from the elite in equilibrium is

$$(8) \quad \hat{e}^*(\beta) := \min \left\{ 1 - \frac{1 - \delta}{\delta(1 - \beta)}, \frac{1}{2} \right\}$$

and $\hat{e}^(\beta) = 1/2$ if and only if*

$$(9) \quad \beta \leq \frac{3\delta - 2}{\delta} < \frac{2\delta - 1}{\delta}.$$

Figure 4 provides a visual illustration for Lemma 3. As discussed above, in an equilibrium where the ruler delegates non-routine tasks to the elite, lower β allows the ruler to commit to larger credible transfers and induce higher elite effort. Therefore, when $\beta \leq \frac{3\delta - 2}{\delta}$, the ruler-optimal level of elite effort $\hat{e}^* = \frac{1}{2}$ can be induced in equilibrium; when $\beta \in (\frac{3\delta - 2}{\delta}, \frac{2\delta - 1}{\delta})$, a positive but inferior effort level $\hat{e}^* \in (0, \frac{1}{2})$ can be induced; and when $\beta \geq \frac{2\delta - 1}{\delta}$, no positive effort can be induced.

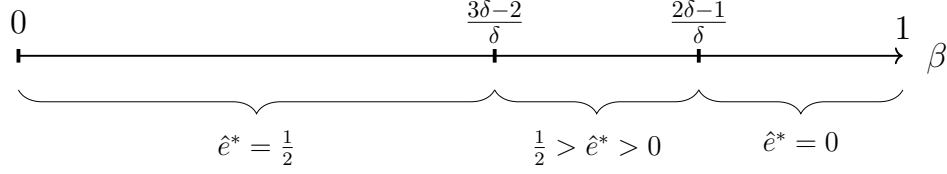


Figure 4: Elite effort as a function of routinization, β .

7. ARISTOCRATIC, BUREAUCRATIC, AND HYBRID REGIMES

The equilibria derived above imply different allocations of routine and non-routine tasks. This section selects the ruler's optimal equilibrium by comparing her payoffs across all the sustainable equilibria and determines the allocation that maximizes her payoff for different ranges of β . These allocations give rise to three types of regimes. A regime is *aristocratic* if, in the ruler-optimal equilibrium, the ruler assigns all tasks to the elite and induces positive effort from the elite; *bureaucratic* if, in the ruler-optimal equilibrium, the ruler assigns routine tasks to the bureaucrat and induces positive effort from the bureaucrat, while leaving non-routine tasks idle and inducing zero effort on them⁷; and *hybrid* if, in the ruler-optimal equilibrium, the ruler assigns routine tasks to the bureaucrat and non-routine tasks to the elite and induces positive effort from both. According to Lemmas 1–3, equilibrium selection depends crucially on the level of routinization, β , as the inducible effort from both agents varies systematically with β .

7.1. Aristocratic Regime

First, consider the case in which routinization is extremely low, so that

$$\beta \leq \frac{1-\delta}{2\delta}.$$

In this range, the ruler is unable to induce any positive effort from the bureaucrat on routine tasks (Lemma 1). This is because the continuation value of future routine cooperation is negligible, so the ruler cannot credibly commit to a sufficiently large transfer. Thus, the ruler has no other choice but to adopt the aristocratic regime. Figure 5 illustrates this regime choice. The ruler delegates all tasks to the elite, and the elite exerts the ruler-optimal effort, $\hat{e} = \frac{1}{2}$, on these tasks, delivering an expected payoff of $\frac{1}{4}$ to the ruler (Lemma 2).

⁷The ruler is indifferent between leaving non-routine tasks idle and assigning them to the bureaucrat, as both induce zero effort.

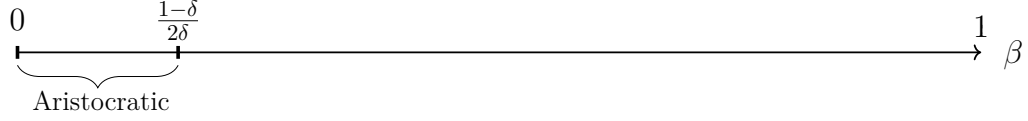


Figure 5: Regime type as a function of routinization, β .

Second, consider the case in which routinization is low but not extremely low, so that

$$\frac{1-\delta}{2\delta} < \beta \leq \frac{1-\delta}{\delta}.$$

In this range, the ruler can induce positive effort from both agents under either the aristocratic or the hybrid regime (Lemmas 2-3), so both regimes are viable and the ruler chooses between them. The two options are equivalent with respect to non-routine tasks: in either case, the ruler assigns non-routine tasks to the elite, induces the ruler-optimal effort $\hat{e} = \frac{1}{2}$, and obtains an expected payoff of $\frac{1}{4}$. The ruler can still induce this ruler-optimal elite effort on non-routine tasks in the hybrid regime because β is not yet high enough to reduce the continuation value of non-routine cooperation to a level that undermines her commitment to the required transfer. The difference lies in routine tasks. In the aristocratic regime, the ruler assigns routine tasks to the elite, induces the ruler-optimal elite effort, $\hat{e} = \frac{1}{2}$, and obtains a payoff of $\frac{1}{4}$ (Lemma 2). In the hybrid regime, she assigns routine tasks to the bureaucrat, induces a positive but not ruler-optimal bureaucratic effort, $\tilde{e}(\beta) \in (0, 1)$, at a cost of $\tilde{e}(\beta)/2$ (Lemma 1). The ruler cannot induce the ruler-optimal bureaucratic effort on routine tasks in the hybrid regime because β remains too low for the continuation value of routine cooperation to sustain the required transfer. A comparison of $\hat{e}(\beta) = \frac{1}{2}$ and $\tilde{e}(\beta) \in (0, 1)$ does not by itself determine the payoff ranking, so I compute the payoffs explicitly. In the hybrid regime, the ruler's per-period payoff from assigning routine tasks to the bureaucrat is at most

$$\tilde{e}(\beta) \left(1 - \frac{\tilde{e}(\beta)}{2} \right).$$

And the ruler chooses to assign routine tasks to the bureaucrat rather than relying solely on the elite for both types of tasks if and only if

$$\tilde{e}(\beta) \left(1 - \frac{\tilde{e}(\beta)}{2} \right) \geq \frac{1}{4},$$

which yields

$$(10) \quad \tilde{e}(\beta) \geq 1 - \frac{\sqrt{2}}{2}.$$

Accordingly, the ruler chooses the hybrid regime only if she can induce a bureaucratic effort higher than $1 - \frac{\sqrt{2}}{2}$. By Lemma 1, the bureaucratic effort that the ruler can induce in equilibrium is bounded above by $2 - \frac{1-\delta}{\delta\beta}$. It follows that the ruler can induce a bureaucratic effort high enough to justify choosing the hybrid regime only when $2 - \frac{1-\delta}{\delta\beta} \geq 1 - \frac{\sqrt{2}}{2}$, that is, when $\beta \geq \frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}$. Hence, within the range $(\frac{1-\delta}{2\delta}, \frac{1-\delta}{\delta}]$, the ruler chooses the aristocratic regime when $\beta \in (\frac{1-\delta}{2\delta}, \frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}]$ and the hybrid regime when $\beta \in (\frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}, \frac{1-\delta}{\delta}]$. Figure 6 illustrates these regime choices.

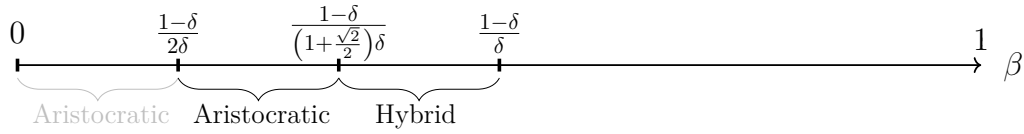


Figure 6: Regime type as a function of routinization, β .

Proposition 2. *In the ruler-optimal equilibrium, the ruler assigns all tasks to the elite and induces positive effort from him if and only if*

$$(11) \quad \beta \leq \frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}.$$

Corollary 1. *If (11) holds, then in the ruler-optimal equilibrium, the ruler assigns all tasks to the elite, makes the transfer of*

$$W^{\mathcal{E}}(y) = \begin{cases} \frac{1}{2}, & y = 1 \\ 0, & \text{otherwise} \end{cases},$$

and gets the payoff of 1/4.

The ruler chooses the aristocratic regime in both $(0, \frac{1-\delta}{2\delta}]$ and $(\frac{1-\delta}{2\delta}, \frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}]$, though for different reasons. In the former, the hybrid regime is not viable on its own, leaving the ruler no choice but to adopt the aristocratic regime. In the latter, both the aristocratic and hybrid regimes are viable on their own, but the ruler still prefers the aristocratic regime. The hybrid regime becomes viable in the latter range because routine tasks arrive slightly more frequently, raising the continuation value of routine cooperation and increasing the cost to the ruler of reneging. This enables her to credibly commit to a larger transfer that

induces a positive, though not ruler-optimal, bureaucratic effort, $\tilde{e}(\beta) \in (0, 1)$. Yet this level of bureaucratic effort remains insufficient to make the hybrid regime optimal for the ruler. Figure 7 depicts the aristocratic regime, shown by the blue segment. Relative to the benchmark shown by the dashed line, the ruler's commitment problem delays the transition to the hybrid regime and postpones the associated gains from bureaucratization.

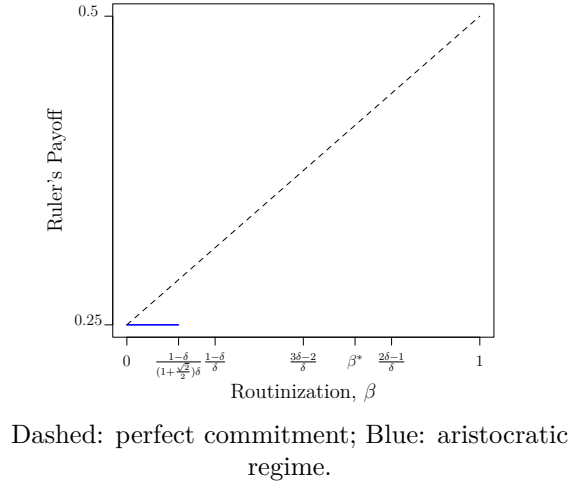


Figure 7: Ruler's payoff as a function of routinization, β .

7.2. Hybrid Regime

Regime selection. As shown above, the ruler chooses the hybrid regime when $\beta \in \left(\frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}, \frac{1-\delta}{\delta} \right]$. I now examine additional ranges of β in which the hybrid regime is optimal. First, consider the case in which routinization is intermediate, though still relatively low, so that

$$\frac{1-\delta}{\delta} < \beta \leq \frac{3\delta-2}{\delta}.$$

In this range, both the aristocratic and hybrid regimes are viable, and the ruler chooses between them. The two options remain equivalent with respect to non-routine tasks: in both regimes, the ruler delegates non-routine tasks to the elite, induces the ruler-optimal elite effort, $\hat{e} = \frac{1}{2}$, and obtains a payoff of $\frac{1}{4}$ (Lemmas 2–3). She can still induce the ruler-optimal elite effort in the hybrid regime because β is not yet high enough to reduce the continuation value of non-routine cooperation to a level that undermines her commitment to the required transfer. The difference lies in routine tasks. In the aristocratic regime, the ruler assigns routine tasks to the elite, induces the ruler-optimal elite effort, $\hat{e} = \frac{1}{2}$, and

obtains a payoff of $\frac{1}{4}$ (Lemma 2). In the hybrid regime, she now assigns routine tasks to the bureaucrat, which allows her to induce the ruler-optimal bureaucratic effort $\tilde{e} = 1$ and obtain a payoff of $\frac{1}{2}$ (Lemma 1). The ruler can now induce the ruler-optimal bureaucratic effort on routine tasks because β is high enough that routine cooperation generates sufficient continuation value to sustain the required transfer. Since $\frac{1}{2} > \frac{1}{4}$, the ruler chooses the hybrid regime. Note that in this hybrid regime, the ruler induces the ruler-optimal level of effort from both agents, just as in the benchmark case, because routinization is at an intermediate level and the ruler expects substantial gains from future cooperation with both agents. Thus, she can credibly commit to the transfers required to induce the optimal level of effort from both agents. Figure 8 illustrates this regime choice.

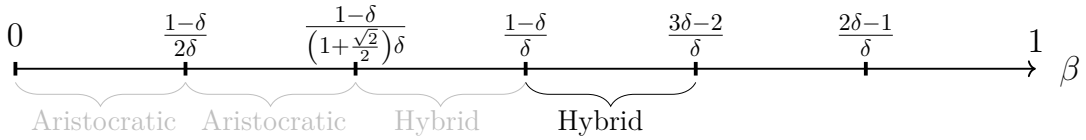


Figure 8: Regime type as a function of routinization, β .

Second, consider the case in which routinization is intermediate, but relatively high, so that

$$\frac{3\delta - 2}{\delta} < \beta \leq \frac{2\delta - 1}{\delta}.$$

In this range, both the aristocratic and hybrid regimes are viable, and the ruler chooses between them. The hybrid regime still has an advantage with respect to routine tasks. In the hybrid regime, the ruler induces the ruler-optimal bureaucratic effort, $\tilde{e} = 1$, on routine tasks and obtains a payoff of $\frac{1}{2}$, whereas in the aristocratic regime, the ruler induces the ruler-optimal elite effort, $\hat{e} = \frac{1}{2}$, on routine tasks and obtains a payoff of $\frac{1}{4}$. Clearly, $\frac{1}{2} > \frac{1}{4}$. The ruler can still induce the ruler-optimal bureaucratic effort on routine tasks in the hybrid regime because β is higher, which strengthens the ruler's commitment to the bureaucrat. However, the aristocratic regime now has an advantage with respect to non-routine tasks. In the aristocratic regime, the ruler assigns non-routine tasks to the elite, induces the ruler-optimal elite effort, $\hat{e} = \frac{1}{2}$, and obtains a payoff of $\frac{1}{4}$ (Lemma 2). In the hybrid regime, the ruler assigns non-routine tasks to the elite but induces only a positive, not ruler-optimal elite effort, $\hat{e}(\beta) \in (0, \frac{1}{2})$ at a cost of $\hat{e}(\beta)$. The ruler can no longer induce the ruler-optimal elite effort on non-routine tasks in the hybrid regime because β becomes sufficiently high that it reduces the continuation value of non-routine cooperation below the level required to sustain the transfer. Therefore, the ruler faces a trade-off between gains from routine tasks in the hybrid regime and gains from non-routine tasks in the aristocratic regime, and the former

exceed the latter. Accordingly, the ruler chooses the hybrid regime. Figure 8 illustrates this regime choice

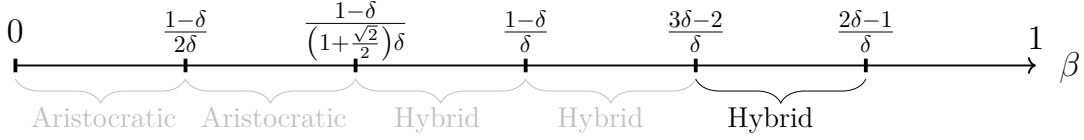


Figure 9: Regime type as a function of routinization, β .

Proposition 3. *In the ruler-optimal equilibrium, the ruler assigns routine tasks to the bureaucrat and non-routine tasks to the elite and induces positive effort from both if and only if*

$$(12) \quad \frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta} < \beta \leq \frac{2\delta-1}{\delta}.$$

In the ruler-optimal equilibrium, the bureaucrat and the elite exert the same levels of effort as if the ruler has perfect commitment if and only if

$$(13) \quad \frac{1-\delta}{\delta} \leq \beta \leq \frac{3\delta-2}{\delta}.$$

Ruler's payoff. The ruler chooses the hybrid regime when $\beta \in \left(\frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}, \frac{2\delta-1}{\delta}\right]$, thereby beginning to realize payoff gains from bureaucratization. This range can be partitioned into three intervals, namely $\left(\frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}, \frac{1-\delta}{\delta}\right)$, $\left(\frac{1-\delta}{\delta}, \frac{3\delta-2}{\delta}\right)$, and $\left(\frac{3\delta-2}{\delta}, \frac{2\delta-1}{\delta}\right)$, each associated with distinct implications for the ruler's payoffs. In what follows, I characterize the mathematical properties of the ruler's payoff across these intervals in turn. Recall that $\frac{3}{4} < \beta < 1$. I begin by deriving the general expression for the ruler's payoff under the hybrid regime.

Corollary 2. *If (12) holds, then in the ruler optimal equilibrium,*

- *if the task is non-routine, $\theta = 0$, the ruler assigns the task to the elite, the elite exerts the effort of $\hat{e}^*(\beta)$, and the ruler makes the transfer of*

$$(14) \quad W^{\mathcal{E}}(y) = \begin{cases} \hat{e}^*(\beta), & y = 1 \\ 0, & \text{otherwise} \end{cases};$$

- *if the task is routine, $\theta = 1$, the ruler assigns the task to the bureaucrat, the bureaucrat*

exerts the effort of $\tilde{e}^*(\beta)$, and the ruler makes the transfer of

$$(15) \quad W^B(y, e) = \begin{cases} \frac{\tilde{e}^*(\beta)}{2}, & y = 1, e = \tilde{e}^*(\beta) \\ 0, & \text{otherwise} \end{cases};$$

- the ruler gets the payoff of

$$(16) \quad (1 - \beta)\hat{e}^*(\beta)(1 - \tilde{e}^*(\beta)) + \beta\tilde{e}^*(\beta)\left(1 - \frac{\tilde{e}^*(\beta)}{2}\right).$$

Interval 1: $\beta \in \left(\frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}, \frac{1-\delta}{\delta}\right)$. According to (3) and (8), $\tilde{e}^*(\beta) = 2 - \frac{1-\delta}{\delta\beta}$ and $\hat{e}^*(\beta) = \frac{1}{2}$ in this interval. Substituting these expressions into (16) gives the ruler's payoff as a function of β . Differentiating this expression with respect to β once and twice yields

$$\left((1 - \beta)\frac{1}{4} + \beta\left(2 - \frac{1 - \delta}{\delta\beta}\right)\frac{1 - \delta}{2\delta\beta}\right)' > 0,$$

and

$$\left((1 - \beta)\frac{1}{4} + \beta\left(2 - \frac{1 - \delta}{\delta\beta}\right)\frac{1 - \delta}{2\delta\beta}\right)'' = -\frac{(1 - \delta)^2}{\delta^2\beta^3} < 0.$$

This indicates that the ruler's payoff is increasing and concave in this interval.

Hence, the ruler induces a positive but not ruler-optimal bureaucratic effort on routine tasks, $\tilde{e} \in (0, 1)$, and the ruler-optimal elite effort, $\hat{e} = \frac{1}{2}$, on non-routine tasks. As a result, further routinization raises the ruler's payoff through two channels. First, higher routinization means that a larger share of tasks can be reallocated to the bureaucrat, who handles them more efficiently. Second, it leads to higher bureaucratic effort, thereby creating an even larger efficiency gap between the two agents. This effect is concave, with diminishing marginal gains as bureaucratic effort gradually approaches its upper limit, $\tilde{e}^*(\beta) = 1$. Taken together, these two effects generate the steepest nonlinear increase in the ruler's payoff across the three regions.

Interval 2: $\beta \in \left(\frac{1-\delta}{\delta}, \frac{3\delta-2}{\delta}\right)$. According to (3) and (8), $\tilde{e}^*(\beta) = 1$ and $\hat{e}^*(\beta) = \frac{1}{2}$. Substituting these expressions into (16) gives the ruler's payoff as a function of β . Differentiating this expression with respect to β yields

$$\left((1 - \beta)\frac{1}{4} + \beta\frac{1}{2}\right)' = \left((1 + \beta)\frac{1}{4}\right)' = \frac{1}{4}.$$

This indicates that the ruler's payoff increases linearly, with a slope that coincides with that

in the benchmark case. This is because the ruler induces the ruler-optimal level of effort from both agents, which cannot be further increased. As a result, higher routinization affects the ruler's payoff only through task reallocation from the elite to the bureaucrat, generating an increase in the ruler's payoff that is proportional to the level of routinization.

Interval 3: $\beta \in \left(\frac{3\delta-2}{\delta}, \frac{2\delta-1}{\delta}\right)$. According to (3) and (8), $\tilde{e}^*(\beta) = 1$ and $\hat{e}^*(\beta) = 1 - \frac{1-\delta}{\delta(1-\beta)}$. Substituting these expressions into (16) gives the ruler's payoff as a function of β , that is,

$$(17) \quad (1-\beta) \left(1 - \frac{1-\delta}{\delta(1-\beta)}\right) \frac{1-\delta}{\delta(1-\beta)} + \beta \frac{1}{2}.$$

Differentiating (17) with respect to β twice yields

$$\left((1-\beta) \left(1 - \frac{1-\delta}{\delta(1-\beta)}\right) \frac{1-\delta}{\delta(1-\beta)} + \beta \frac{1}{2} \right)'' = -\frac{2(1-\delta)^2}{\delta^2(1-\beta)^3} < 0,$$

which indicates that the ruler's payoff is strictly concave in this region. Differentiating (17) with respect to β once and setting the derivative equal to zero,

$$\left((1-\beta) \left(1 - \frac{1-\delta}{\delta(1-\beta)}\right) \frac{1-\delta}{\delta(1-\beta)} + \beta \frac{1}{2} \right)' = 0,$$

yields

$$\beta^* = \frac{(1+\sqrt{2})\delta - \sqrt{2}}{\delta} \in \left(\frac{3\delta-2}{\delta}, \frac{2\delta-1}{\delta}\right).$$

This is the critical point at which the derivative of the ruler's payoff with respect to routinization changes sign from positive to negative. The ruler's payoff therefore follows an inverted-U shape in this region, attaining its maximum at the critical point and its minimum at one of the endpoints. To establish that the ruler strictly prefers the hybrid regime throughout this region, it suffices to verify that the payoff at both endpoints exceeds that under the aristocratic regime. At the right endpoint, $\beta = \frac{2\delta-1}{\delta}$, the ruler's payoff under the hybrid regime exceeds that under the aristocratic regime, $\frac{1}{4}$, as

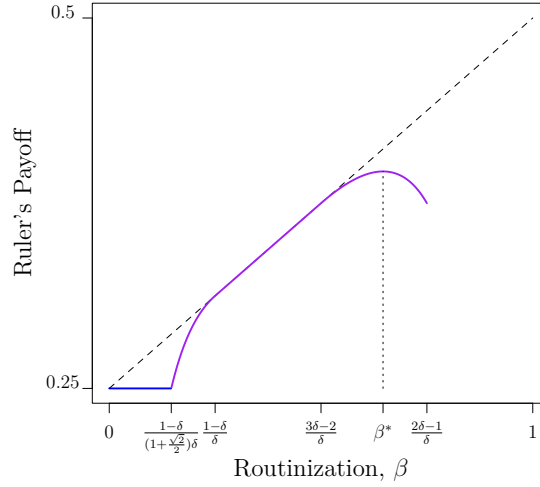
$$\left(1 - \frac{2\delta-1}{\delta}\right) \left(1 - \frac{1-\delta}{\delta(1-\frac{2\delta-1}{\delta})}\right) \frac{1-\delta}{\delta(1-\frac{2\delta-1}{\delta})} + \frac{2\delta-1}{2\delta} > \frac{1}{4}$$

for all $\delta \in [\frac{3}{4}, 1)$. The left endpoint has already been verified in Interval 2. Hence, throughout this region, the ruler strictly prefers the hybrid regime to the aristocratic regime.

In sum, the ruler induces the ruler-optimal bureaucratic effort on routine tasks, $\tilde{e} = 1$,

and a positive but not ruler-optimal elite effort, $\hat{e} \in (0, \frac{1}{2})$, on non-routine tasks. As a result, further routinization affects the ruler's payoff through two opposing channels. On the one hand, reallocating a larger share of tasks to the bureaucrat raises the ruler's payoff on routine tasks. On the other hand, elite effort on non-routine tasks declines. Importantly, elite effort on non-routine tasks begins to decline only in this interval. Although the continuation value of non-routine cooperation continues to decline with a rising β , elite effort is not affected in the previous intervals because the continuation value, though declining, still remains above the threshold required to sustain the ruler-optimal level of elite effort. However, once the continuation value falls below this threshold, a higher β begins to reduce elite effort. Hence, the ruler's payoff reflects a tension between rising efficiency in routine tasks and declining efficiency in non-routine tasks, and the former initially dominates but is eventually outweighed by the latter. As elite effort eventually falls to zero at the right endpoint of the interval, $\beta = \frac{2\delta-1}{\delta}$, the ruler's payoff also falls to its local minimum.

The Illustration. Figure 10 depicts the hybrid regime, shown by the purple segment. It exhibits a concave rise, followed by a linear increase, and then an inverted-U shape. The linear portion overlaps with the benchmark, shown by the dashed line. The slope changes sign at the critical point β^* . It reflects the central idea of this paper: although initial bureaucratization yields substantial gains, further bureaucratization weakens elite incentives and ultimately reduces the ruler's payoff.



Dashed: perfect commitment; Blue: aristocratic regime.
Purple: hybrid regime.

Figure 10: Ruler's payoff as a function of routinization, β .

7.3. Bureaucratic Regime

Consider the case in which routinization is very high, so that

$$\beta \geq \frac{2\delta - 1}{\delta}.$$

In this range, the ruler is unable to induce positive elite effort if she delegates only non-routine tasks to the elite (Lemma 3), so the hybrid regime is not viable. The ruler has two options. One option is the aristocratic regime, in which she assigns all tasks to the elite, induces the ruler-optimal elite effort, $\hat{e}^*(\beta) = \frac{1}{2}$, and obtains a payoff of $\frac{1}{4}$ (Lemma 2). The other option is the bureaucratic regime, in which she delegates routine tasks to the bureaucrat and induces the ruler-optimal bureaucratic effort, $\tilde{e}^*(\beta) = 1$, while leaving non-routine tasks idle (Lemma 1). In the latter case, the resulting ruler's payoff is

$$\beta \tilde{e}^*(\beta) \left(1 - \frac{\tilde{e}^*(\beta)}{2}\right) = \beta \left(1 - \frac{1}{2}\right) = \frac{\beta}{2}.$$

Note that because

$$\beta \geq \frac{2\delta - 1}{\delta} > \frac{1}{2},$$

it follows that $\beta/2 > 1/4$. Hence, the ruler strictly prefers the bureaucratic regime over the aristocratic regime and willingly gives up the non-routine tasks altogether. Figure 11 illustrates this regime choice.

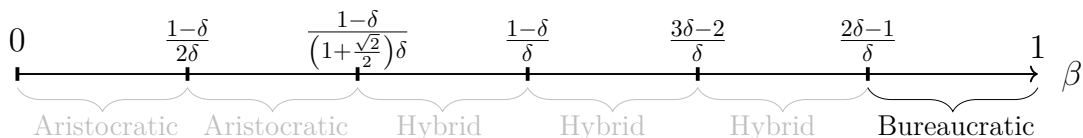


Figure 11: Regime type as a function of routinization, β .

Proposition 4. *If*

$$(18) \quad \beta \geq \frac{2\delta - 1}{\delta},$$

then in the ruler-optimal equilibrium, the ruler assigns routine tasks to the bureaucrat and induces positive effort from it, while she induces no effort on non-routine tasks.

When routinization is very high, non-routine tasks arrive so infrequently that delegating only non-routine tasks to the elite yields a continuation value that is too low to sustain

a transfer large enough to induce any positive elite effort. To induce positive elite effort, the ruler must assign all tasks to the elite, in which case she induces the ruler-optimal elite effort $\hat{e} = \frac{1}{2}$, yielding a payoff of $\frac{1}{4}$. Alternatively, the ruler can cooperate exclusively with the bureaucrat, as routine tasks arrive frequently enough that delegating them to the bureaucrat generates a continuation value sufficient to sustain the transfer needed to induce the ruler-optimal bureaucratic effort $\tilde{e} = 1$, yielding a payoff of $\frac{1}{2}$. In sum, the ruler can either earn a lower payoff on all tasks in the aristocratic regime or earn a higher payoff on exclusively routine tasks in the bureaucratic regime. Given the predominant share of routine tasks at high levels of routinization, the ruler strictly prefers the bureaucratic regime ($\frac{1}{2} > \frac{1}{4}$). In this regime, she no longer cooperates with the elite, even though the elite remains the only agent capable of handling non-routine tasks.

Corollary 3. *If (18) holds, then in the ruler-optimal equilibrium, the ruler assigns routine tasks to the bureaucracy, makes the transfer of*

$$W^B(y, e) = \begin{cases} \frac{1}{2}, & y = 1, e = 1 \\ 0, & \text{otherwise} \end{cases},$$

and gets the payoff of $\beta/2$.

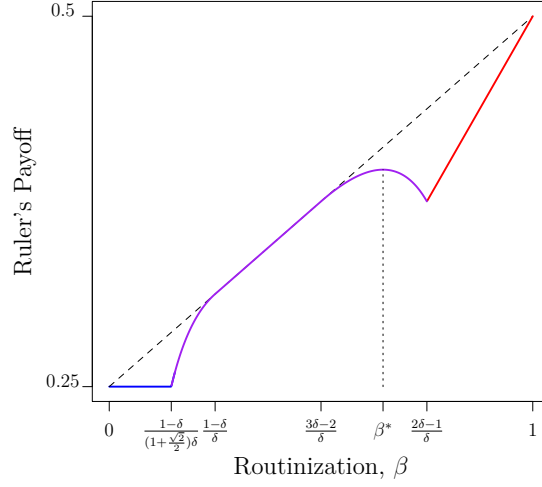
Figure 12 depicts the bureaucratic regime, shown by the red segment. It exhibits a linear increase in the ruler's payoff with respect to routinization, with a steeper slope than in the benchmark case. This is because, in the bureaucratic regime, the ruler's payoff increases as the bureaucrat takes on previously idle non-routine tasks, rather than reallocating tasks away from the elite, as in the benchmark case.

7.4. The ruler's maximal payoff

To summarize, let $u(\beta)$ denote the ruler's payoff in the ruler-optimal equilibrium. According to the above analysis,

$$(19) \quad u(\beta) = \begin{cases} \frac{1}{4}, & \beta \leq \frac{1-\delta}{\left(1+\frac{\sqrt{2}}{2}\right)\delta} \\ (1-\beta)\frac{1}{4} + \beta\left(2 - \frac{1-\delta}{\delta\beta}\right)\frac{1-\delta}{2\delta\beta}, & \frac{1-\delta}{\left(1+\frac{\sqrt{2}}{2}\right)\delta} < \beta < \frac{1-\delta}{\delta} \\ (1-\beta)\frac{1}{4} + \beta\frac{1}{2}, & \frac{1-\delta}{\delta} \leq \beta \leq \frac{3\delta-2}{\delta} \\ (1-\beta)\left(1 - \frac{1-\delta}{\delta(1-\beta)}\right)\frac{1-\delta}{\delta(1-\beta)} + \beta\frac{1}{2}, & \frac{3\delta-2}{\delta} < \beta < \frac{2\delta-1}{\delta} \\ \beta\frac{1}{2}, & \beta \geq \frac{2\delta-1}{\delta} \end{cases}.$$

Proposition 5. *$u(\beta)$ is continuous in β and it is*



Dashed: perfect commitment; Blue: aristocratic regime;
 Purple: hybrid regime; Red: bureaucratic regime.

Figure 12: Ruler's payoff as a function of routinization, β .

- *constant in β if $\beta \leq (1 - \delta) / \left((1 + \sqrt{2}/2)\delta \right)$;*
- *strictly increasing in β if $(1 - \delta) / \left((1 + \sqrt{2}/2)\delta \right) < \beta < \beta^*$, where*

$$\beta^* := \frac{(1 + \sqrt{2})\delta - \sqrt{2}}{\delta} \in \left(\frac{3\delta - 2}{\delta}, \frac{2\delta - 1}{\delta} \right);$$

- *attains local maximum at $\beta = \beta^*$;*
- *strictly decreasing in β if $\beta^* \leq \beta < (2\delta - 1)/\delta$;*
- *strictly increasing in β if $\beta \geq (2\delta - 1)/\delta$.*
- *attains global maximum at $\beta = 1$.*

Several comments can be made regarding this summary of results. First, this pattern confirms the central claim of the paper that even transitioning to a perfectly efficient bureaucracy leads the ruler's payoff to trace a bumpy journey, which departs at two places from the linear benchmark. The first departure occurs at the outset, when the ruler's payoff plateaus as bureaucratization is delayed: The second occurs at the intermediate stage, when the ruler's payoff declines even as bureaucratization continues to advance. The two departures arise from analogous mechanisms operating in opposite directions: when β is low, elites remain sufficiently effective to undermine the ruler's developing commitment

to bureaucrats; when β is high, bureaucrats become sufficiently effective to undermine the ruler's remaining commitment to elites. Therefore, the hybrid regime is neither profitable at low levels of routinization nor sustainable at high levels of routinization.

Second, when the ruler is in the region $\beta \in [\beta^*, \beta^\dagger]$, where $\beta^\dagger = 2\sqrt{2} - 1 + \frac{2(1-\sqrt{2})}{\delta}$ and $u(\beta^*) = u(\beta^\dagger)$, the ruler would be better off reverting to lower levels of routinization if she could choose (see Figure 13), though modeling routinization as an endogenous choice lies outside the scope of this paper.

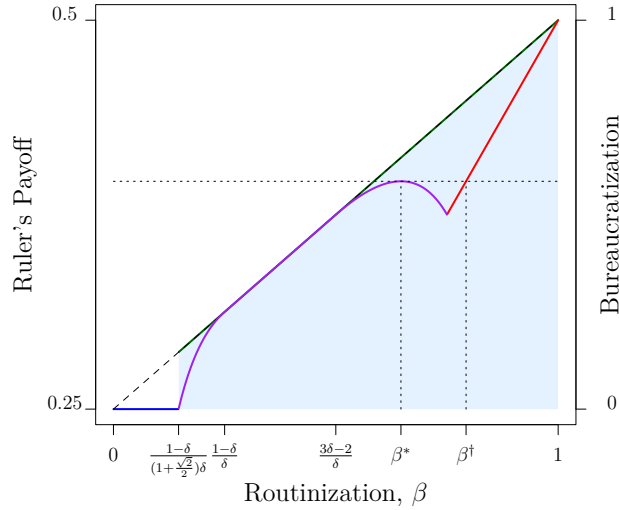
Third, the decline would be more severe if the bureaucracy is subject to moral hazard itself. However, as long as bureaucratic moral hazard remains marginally lower than that of elites, the transition toward bureaucratic governance still occurs in equilibrium, and the N-shaped pattern is preserved. Bureaucratic inefficiencies would nonetheless lengthen the initial delay to bureaucratization and deepen the decline (see the extension).

Fourth, under the assumption of a perfectly efficient bureaucracy, the upward trend of the red line appears particularly steep, and the global maximum occurs under full routinization, that is, when $\beta = 1$. Two caveats are worth noting. First, if the bureaucracy is itself subject to moral hazard, the upward slope would be more attenuated, and the peak at the global maximum would be less pronounced. Second, full bureaucratization is unlikely to happen in history, as there are always domains that are difficult to monitor and reward effectively in practice; and if β is below $\beta^\dagger$, the ruler's payoff is worse than local maximum at β^* .

Fifth, bureaucratization is defined as the share of tasks delegated to bureaucrats in equilibrium, which is also a function of routinization, as shown by the green line and the shaded area beneath it (see Figure 13). Aside from an initial delay, bureaucratization largely progresses in tandem with routinization, as the ruler delegates all routinized tasks to bureaucrats in both the hybrid and bureaucratic regimes. The central message of the paper is reinforced here: while the transition to a more efficient agent can be smooth once underway, it may not necessarily be always beneficial to the ruler.

Sixth, total surplus, defined as the aggregate payoff of the ruler, the elite, and the bureaucrat, also follows an N-shaped pattern as a function of routinization (see Figure 14). This indicates that the model's results extend to a broader notion of state capacity beyond that of the central government. The total surplus curve displays a more pronounced mid-stage decline because the reduction in the surplus generated by elite effort (hereafter, elite surplus) is now directly incorporated into the aggregate payoff. Relative to the ruler's payoff curve, two differences are worth noting. First, total surplus exhibits a dip upon entry into the hybrid regime at $\beta = \frac{1-\delta}{(1+\frac{\sqrt{2}}{2})\delta}$. This is because bureaucratic effort at this stage is still lower than that of the elite, so while the ruler might find this transition lucrative, as she captures a larger share from bureaucratic effort than from elite effort,

total surplus nonetheless declines. Second, the slope of total surplus changes sign at $\beta = \frac{3\delta-2}{\delta}$, earlier than β^* . At this threshold, the ruler's commitment to the elite begins to weaken, causing a decline in elite surplus, which is now directly incorporated into total surplus. More precisely, the resulting drop in elite surplus is nonlinear and first-order, whereas the gains in total surplus from bureaucratization are linear and second-order, implying an overall decline in total surplus.

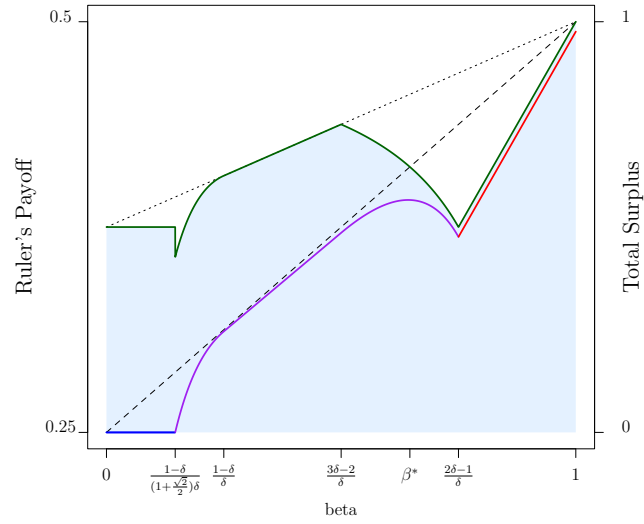


Dashed: perfect commitment; Blue: aristocratic regime;
 Purple: hybrid regime; Red: bureaucratic regime;
 Green with Shaded Area: Bureaucratization.

Figure 13: Ruler's payoff and bureaucratization as a function of routinization, β .

8. CONCLUSION

History is difficult to disentangle, as it unfolds through numerous contingencies and reversals of fortune. In the eighteenth century, a wide range of state forms coexisted, each on a trajectory that was only beginning to unfold or nearing completion, yet remained difficult to interpret. These included the highly aristocratic “noble republic” of the Polish–Lithuanian Commonwealth, marked by weak state capacity and growing vulnerability to more centralized neighbors such as Prussia, Austria, and Russia; the rising fiscal-military state of Great Britain, where cooperative elites and an increasingly professional bureaucracy supported the early stages of global expansion; and the highly bureaucratic Qing state in China, which had once been a dominant power but was beginning to face mounting internal and external pressures. From the vantage point of the



Dashed: perfect commitment benchmark for ruler's payoff;
Dotted: perfect commitment benchmark for total surplus;
Blue: aristocratic regime; Purple: hybrid regime;
Red: bureaucratic regime;
Green with Shaded Area: Total Surplus.

Figure 14: Ruler's payoff and total surplus as a function of routinization, β .

time, it was far from clear why these states had taken such divergent paths or what their futures would hold.

With the benefit of several centuries of hindsight, the broad contours of these trajectories have become clearer. The Polish–Lithuanian Commonwealth collapsed and was partitioned; Great Britain rose rapidly to global prominence before settling into a position of diminished but enduring influence; and the Qing state fell, followed much later by the resurgence of a modern bureaucratic Chinese state. Viewed in retrospect, these experiences offer useful clues to the underlying logic of historical transformation.

The inability of certain states to transform themselves in the face of political difficulties suggests that this process is, at least in part, exogenous over short historical horizons. From a later perspective, one might observe that the Polish–Lithuanian Commonwealth could have benefited from greater centralization, while the Qing state, along with other highly bureaucratized empires such as the Ottoman Empire, might have fared better with less rigid administrative structures. Yet such adjustments were not within their reach, and each was ultimately drawn along a path that led to a difficult and often tragic end.

If the process of institutional regularization and personnel bureaucratization indeed entails a turbulent path, in which gains in state capacity are followed by an endogenous period of decline, then the apparent failures of states may be less a reflection of poor judgment than of structural constraint. The strategic responses of rulers, elites, and bureaucrats to one another make such dynamics difficult to escape. In this sense, greater sympathy may be warranted toward those who struggled unsuccessfully against adverse historical currents, and a more measured appraisal is warranted for those who happened to govern during favorable structural trends. Differences in institutional quality or elite composition may shift the timing or magnitude of the rise and the downturn, but they do not fundamentally alter the trajectory itself.

The establishment and maintenance of a highly centralized bureaucratic state are therefore a demanding and precarious undertaking. Many polities have embarked upon this path only to falter during the difficult middle phase, leaving successors unable to resume the process. In the present, however, we may be observing an unusually large-scale attempt in China to push bureaucratic development further along this trajectory, probably through sustained technological innovation and administrative refinement. Whether such an effort can ultimately overcome the inherent vulnerabilities of this trajectory and avoid the fate that befell earlier states remains an open question.

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